Structuri liniare LAB 2 (AA) 14 oct 2020

①

i=1	c_1	1
while(v[i]>=0){	c_2	t
if(v[i]%2==0)	c_3	t-1
print(v[i]);	c_4	p
i++;	c_5	t-1

}

- toate operațiile $T(n) = 1c_1 + tc_2 + (t-1)c_3 + pc_4 + (t-1)c_5$
- cazul cel mai favorabil $t=1$ $p=0$

$T(n) = c_1 + c_2 = \Theta(1)$

- cazul cel mai defavorabil $t=n$ $p=n-1$

$T(n) = c_1 + nc_2 + (n-1)c_3 + (n-1)c_4 + (n-1)c_5$

$= n(c_2+c_3+c_4+c_5) + (c_1 - c_3 - c_4 - c_5) = \Theta(n)$

- cazul mediu $t = h_1 + (1-h_1)n$ $p = h_2 \cdot 0 + (1-h_2)(n-1)$

$T(n) = c_1 + (h_1 + (1-h_1)n)c_2 + (h_1 + (1-h_1)n)c_3 + ((1-h_2)(n-1))c_4 + (h_1 + (1-h_1)n)c_5$

$= n((1-h_1)c_2 + (1-h_1)c_3 + (1-h_2)c_4 + (1-h_1)c_5) + (c_1 + h_1c_2 + h_1c_3 - c_3 + (1-h_2)c_4 + h_1c_5 - c_5)$

$= \Theta(n)$

- operații critice

a) c. cel mai favorabil $t=1$ $p=0$ $\Rightarrow$ 1, 2

$T(n) = c_1 + c_2 = \Theta(1)$

b) c. cel mai defavorabil $t=n$ $p=n-1$ $\Rightarrow$ 3, 4, 5

$T(n) = (n-1)c_3 + (n-1)c_4 + (n-1)c_5 = n(c_3+c_4+c_5) - (c_3+c_4+c_5) = \Theta(n)$

c) c. mediu $t = h_1 + (1-h_1)n$ $p = (1-h_2)(n-1)$

$T(n) = (h_1 + (1-h_1)(n-1))c_3 + ((1-h_2)(n-1))c_4 + (h_1 + (1-h_1)(n-1))c_5$

$= n((1-h_1)c_3 + (1-h_2)c_4 + (1-h_1)c_5) + (h_1c_3 - c_3 - (1-h_2)c_4 + h_1c_5 - c_5) =$

$= \Theta(n)$

② a) $n^2+n = O(n^2) \Leftrightarrow \exists c \in \mathbb{R}_+^*, (\exists)\, n_0 \in \mathbb{N}^*$ a.î. $(n^2+n) \le cn^2 \ \forall n \ge n_0$

Alegem $c=2 \Rightarrow n^2+n \le 2n^2$

$n \le n^2$ A $\Rightarrow n \ge 1 \Rightarrow n_0 = 1 \Rightarrow$

$\Rightarrow n^2+n = O(n^2)$

b) $n^2+n \neq \sigma(n^2)$

$\lim\limits_{n\to\infty} \frac{f(n)}{g(n)} = \lim\limits_{n\to\infty} \frac{n^2+n}{n^2} = 1 \neq 0 \Rightarrow n^2+n \neq \sigma(n^2)$

c) $n^3+n^2 = \Theta(n^3) \Leftrightarrow \exists c_1, c_2 \in \mathbb{R}_+^*, \exists\, n_0 \in \mathbb{N}^*$ a.î. $c_1 n^3 \le n^3+n^2 \le c_2 n^3$

Alegem $c_1 = 1$, $c_2 = 2$ $\Rightarrow n^3 \le n^3+n^2 \le 2(n^3+n^2)$

$0 \le n^2$ $\forall n \ge 1$; $0 \le n^3+n^2$ $\forall n \ge 1$ $\Rightarrow n_0 = 1 \in \mathbb{N}^* \Rightarrow n^3+n^2 = \Theta(n^3)$

d) $f(n) = O(g(n)) \Leftrightarrow g(n) = \Omega(f(n))$

$f(n) = O(g(n)) \Leftrightarrow \exists c \in \mathbb{R}_+^*, n_0 \in \mathbb{N}^*$ a.î. $f(n) \le c g(n) \ \forall n \ge n_0 \Leftrightarrow \exists c \in \mathbb{R}_+^*, n_0 \in \mathbb{N}^*$

a.î. $\frac{1}{c} f(n) \le g(n) \ \forall n \ge n_0 \Leftrightarrow \exists c' = \frac{1}{c} \in \mathbb{R}_+^*, (\exists)\, n_0' = n_0 \in \mathbb{N}^*$ a.î.

$c' f(n) \le g(n) \ \forall n \ge n_0' \Leftrightarrow g(n) = \Omega(f(n))$

e) $f(n) = \Theta(g(n)) \Leftrightarrow \begin{cases} f(n) = O(g(n)) \\ f(n) = \Omega(g(n)) \end{cases}$

"$\Rightarrow$" $f(n) = \Theta(g(n)) \Rightarrow \exists c_1, c_2 \in \mathbb{R}_+^*, (\exists)\, n_0 \in \mathbb{N}^*$ a.î.

$c_1 g(n) \le f(n) \le c_2 g(n)$

$\Omega(g(n))$ $O(g(n))$

"$\Leftarrow$" $f(n) = O(g(n)) \Rightarrow \exists c_1 \in \mathbb{R}_+^*, (\exists)\, n_1 \in \mathbb{N}^*$ a.î. $f(n) \le c_1 g(n) \ \forall n_1 \le n$ /$\Rightarrow$

$f(n) = \Omega(g(n)) \Rightarrow \exists c_2 \in \mathbb{R}_+^*, (\exists)\, n_2 \in \mathbb{N}^*$ a.î. $f(n) \ge c_2 g(n) \ \forall n_2 \le n$ /$\Rightarrow$

$\Rightarrow \exists c_1, c_2 \in \mathbb{R}_+^*, (\exists)\, n_0 = \max(n_0, n_2) \in \mathbb{N}^*$ a.î.

$c_2 g(n) \le f(n) \le c_1 g(n) \ \forall n \ge n_0 \Rightarrow f(n) = \Theta(g(n))$

Struțu Ilinca LAB 3 (AA) 21 oct 2020

```
maxim(v,i,j){
    if(i==j)                                   C1
        return v[i];                           C2
    m=(i+j)/2;
    return max(maxim(v,i,m), maxim(v,m+1,j));  } C3
}
```

$i=j \Rightarrow k_1 = C_1 + C_2$

$i \neq j \Rightarrow k_2 = C_1 + C_3$

$$T(m) = \begin{cases} k_1, & m=1 \\ T(\lfloor m/2 \rfloor) + T(\lceil m/2 \rceil) + k_2, & m>1 \end{cases}$$

Iterativ

$$T(m) = \begin{cases} \Theta(1) & m=1 \\ 2T(m/2) + \Theta(1) & m>1 \end{cases}$$

$T(m) = 2T(m/2) + \Theta(1) \quad /2^0$

$T(m/2) = 2T(m/4) + \Theta(1) \quad /2^1$

$\vdots$

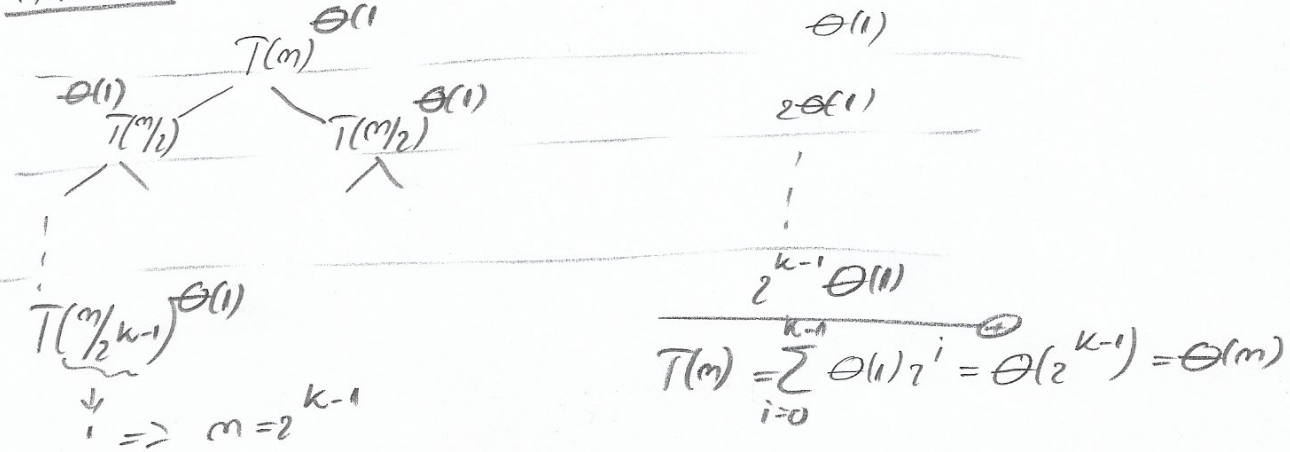
$T(m/2^k) = 2T(m/2^{k+1}) + \Theta(1) \quad /2^k$ unde $m/2^{k+1} \to 1$, $m = 2^{k+1}$

$$T(m) = 2^{k+1} T(m/2^{k+1}) + \sum_{i=0}^{k} 2^i \Theta(1)$$

$T(m/2^{k+1}) \downarrow T(1) = \Theta(1)$

$$T(m) = m\,\Theta(1) + \Theta\left(\sum_{i=0}^{k} 2^i\right) = \Theta(m) + \Theta(m-1) = \Theta(2m-1) = \Theta(m)$$

Arbore



$$T(m) = \sum_{i=0}^{k-1} \Theta(1) 2^i = \Theta(2^{k-1}) = \Theta(m)$$

② cautare(v, i, j, x){ — c_1

if(i==j) — c_2

return i;

m=(i+j)/2; — c_3

if(v[m]>=x)

cautare(v, i, m, x);

else

cautare(v, m+1, j, x);

}

i=j => $k_1 = c_1 + c_2$ daca i≠j => $k_2 = c_1 + c_3$

$$T(m) = \begin{cases} k_1, & m=1 \\ T(\lceil m/2 \rceil) + k_2, & m>1 \end{cases} \Rightarrow \begin{array}{l} T(m) = T(m/2) + \Theta(1) \\ T(1) = \Theta(1) \end{array}$$

Iterativ

$T(m) = T(m/2) + \Theta(1)$

$T(m/2) = T(m/4) + \Theta(1)$

!

!

$T(m/2^k) = T(m/2^{k+1}) + \Theta(1)$ unde $m = 2^{k+1} \Rightarrow k+1 = \lg m$

$T(m) = T(m/2^{k+1}) + (k+1)\Theta(1) = T(1) + \lg m \cdot \Theta(1) = \Theta(1) + \lg(m)\Theta(1) = \Theta(\lg m)$

Substitutie

alegem $T(m) = \Theta(\lg m + 1)$

$\exists c_1, c_2 \in \mathbb{R}_+^*, (\exists) m_0 \in \mathbb{N}^*$ a.i. $c_1(\lg m + 1) \le T_m \le c_2(\lg m + 1)$

caz de baza $m=1 \Rightarrow c_1 \le k_1 \le c_2 \Rightarrow m_0 = 1$

pas $m/2 \to m$

ipoteza $c_1(\lg m/2 + 1) \le T(m/2) \le c_2(\lg m/2 + 1) \mid + k_2$

$c_1(\lg m - 1 + 1) + k_2 \le T(m) \le c_2(\lg m - 1 + 1) + k_2$

$c_1 \lg m + k_2 \le T(m) \le c_2 \lg m + k_2$

$c_1(\lg m + 1) < c_1(\lg m + 1) + (k_2 - c_1) \le T(m) \le c_2(\lg m + 1) + (k_2 - c_2) \le c_2(\lg m + 1)$

($k_2 - c_1 \ge 0$, $k_2 - c_2 \le 0$)

$c_1(\lg m + 1) \le T(m) \le c_2(\lg m + 1) \Rightarrow$

$c_1 \le k_2 \le c_2$

$\Rightarrow T(m) = \Theta(\lg m + 1) \Leftrightarrow \begin{array}{l} c_1 \le k_1 \le c_2 \\ c_1 \le k_2 \le c_2 \end{array} \Rightarrow \exists \begin{array}{l} c_1 = \min(k_1, k_2) \\ c_2 = \max(k_1, k_2) \end{array} \in \mathbb{R}_+^*, \ m_0 = 1$

a.i. $c_1(\lg m + 1) \le T(m) \le c_2(\lg m + 1)$

Structuri liniare LAB 4 (AA) 28 oct 2020

① Rezolvați cu metoda Master

a) $T(n) = 16T(n/4) + n^2$ $\log_b a = n^{\log_4 16} = n^2$ $\Rightarrow f(n) = \Theta(n^2)$ cazul 2

$a = 16$ $b = 4 \Rightarrow n^{\log_b a} = n^{\log_4 16} = n^2$

$f(n) = n^2$

$\Rightarrow T(n) = \Theta(n^2 \lg n)$

b) $T(n) = 4T(n/2) + \Theta(n)$

$a = 4$ $b = 2 \Rightarrow n^{\log_b a} = n^{\lg 4} = n^2$ cazul 1 $\Rightarrow$

$\Rightarrow f(n) = kn = O(n^{2-\varepsilon}) \Leftrightarrow \varepsilon = 1 > 0 \Rightarrow T(n) = \Theta(n^2)$

c) $T(n) = 2T(n/2) + \Theta(n^2)$

$a = 2$ $b = 2 \Rightarrow n^{\log_2 2} = n \Rightarrow$ cazul 3

$\Rightarrow f(n) = kn^2 = \Omega(n^{1+\varepsilon}) \Leftrightarrow \varepsilon = 2 > 0$ $\Rightarrow$

$2 \cdot (n/2)^3 \leq cn^3$

$n^3/4 \leq cn^3 \Rightarrow c = \frac{1}{4}$ $\Rightarrow$

$\Rightarrow T(n) = \Theta(n^3)$

d) $T(n) = 2T(n/2) + n \lg n$

$a = 2$ $b = 2 \Rightarrow n^{\log_b a} = n^1 = n$

cazul 2: $f(n) = \Theta(n) \Leftrightarrow n \lg n = \Theta(n)$ F

cazul 1: $f(n) = n \lg n \neq O(n^{1-\varepsilon})$ F

cazul 3: $f(n) = n \lg n = \Omega(n^{1+\varepsilon})$ $\Rightarrow$ F $\Rightarrow$

$\lim_{\substack{n \to \infty \\ \varepsilon > 0}} \frac{n \lg n}{n^{1+\varepsilon}} = \lim_{\substack{n \to \infty \\ \varepsilon > 0}} \frac{\lg n}{n^{\varepsilon}} = 0$

$\Rightarrow$ nu se poate aplica metoda Master

② a) metoda iterativă

$T(n) = 2T(n/2) + n \lg n \quad /\cdot 2^0$

$T(n/2) = 2T(n/2^2) + \frac{n}{2}\lg\frac{n}{2} \quad /\cdot 2$

⋮

$T(n/2^k) = 2T(n/2^{k+1}) + \frac{n}{2^k}\lg\frac{n}{2^k} \quad /\cdot 2^k$ unde $2^{k+1} = n$, $k+1 = \lg n$

$T(n) = 2^{k+1}T(n/2^{k+1}) + \sum_{i=0}^{k} n\lg\frac{n}{2^i}$

$= nT(1) + \sum_{i=0}^{k} n\lg\frac{n}{2^i} = nT(1) + \sum_{0}^{k}(\lg n - i\lg 2)n$

$= nT(1) + (k+1)n\lg n - n\sum_{0}^{k} i = n\lg n\lg n - n\frac{\lg n\cdot\lg(n-1)}{2}$

$= n\lg^2 n - n\frac{\lg^2 n}{2} + \frac{n\lg n}{2} = \frac{n\lg^2 n}{2} + \frac{n\lg n}{2} = \frac{n\lg n(\lg n+1)}{2}$

$T(n) = \Theta(n\lg^2 n)$

b) $T(n) = 2T(n-1) + \Theta(1)$

metoda iterativă

$T(n) = 2T(n-1) + 1 \quad /\cdot 2^0$

$T(n-1) = 2T(n-2) + 1 \quad /\cdot 2$

⋮

$T(n-k) = 2T(n-k-1) + 1 \quad /\cdot 2^k$ unde $n-k-1 = 1$, $k+1 = n-1$

$T(n) = 2^{k+1}T(1) + \sum_{0}^{k} 2^i$

$T(n) = 2^{n-1}T(1) + (2^{n-1} - 1)$

$= 2^{n-1}\Theta(1) + 2^{n-1} - 1 = 2^n - 1 = \Theta(2^n - 1)$

① $\forall l_1$ $concat(l_1, nil) = l_1$

- caz de bază

$l_1 = nil \Rightarrow concat(l_1, nil) = concat(nil, nil) = nil = l_1$

- ip de ind $concat(l_1, nil) = l_1$
- $concat(cons(x, l_1), nil) = cons(x, l_1)$

$concat(cons(x, l_1), nil) \overset{concat}{=} cons(x, \underbrace{concat(l_1, nil)}_{\text{ip ind}}) = cons(x, l_1)$

② $\forall l_1, l_2, l_3 \Rightarrow concat(l_1, concat(l_2, l_3)) = concat(concat(l_1, l_2), l_3)$

- caz de bază $l_1 = nil \Rightarrow concat(nil, concat(l_2, l_3)) = concat(l_2, l_3) =$

$= concat(concat(nil, l_2), l_3) = concat(concat(l_1, l_2), l_3)$

- ip de ind $concat(l, concat(l_2, l_3)) = concat(concat(l, l_2), l_3)$
- dem pt. $l_1 = cons(x, l)$ $concat(cons(x, l), concat(l_2, l_3)) = concat(concat(cons(x, l), l_2), l_3)$

$concat(cons(x, l), concat(l_2, l_3)) \overset{concat}{=} cons(x, \underbrace{concat(l, concat(l_2, l_3))}_{\text{ip ind}}) =$

$= cons(x, concat(concat(l, l_2), l_3)) \overset{concat}{=} concat(cons(x, concat(l, l_2)), l_3) \overset{concat}{=}$

$= concat(concat(cons(x, l), l_2), l_3)$

3.

③ $\forall l_1, l_2$, $rev(concat(l_1, l_2)) = concat(rev(l_2), rev(l_1))$

• caz de bază $l_1 = nil \Rightarrow rev(concat(nil, l_2)) \overset{rev}{=} rev(l_2) = concat(rev(l_2), nil) = concat(rev(l_2), rev(nil))$

• ip de ind. $rev(concat(l, l_2)) = concat(rev(l_2), rev(l))$

$l_1 = cons(x, l)$

• arătăm că $rev(concat(cons(x, l), l_2)) = concat(rev(l_2), rev(cons(x, l)))$

$rev(concat(cons(x, l), l_2)) \overset{concat}{=}$

$= rev(cons(x, concat(l, l_2))) \overset{rev}{=}$

$= concat(rev(concat(l, l_2)), cons(x, nil)) \overset{ip.\ ind}{=}$

$= concat(concat(rev(l_2), rev(l)), cons(x, nil)) \overset{ex2}{=}$

$= concat(rev(l_2), concat(rev(l), cons(x, nil))) \overset{rev}{=}$

$= concat(rev(l_2), rev(cons(x, l)))$

P1 n...1

3.

(4) $\forall l_1, l_2$ length(concat(l_1, l_2)) = length(l_1) + length(l_2)

- caz do bază $l_1 = nil \Rightarrow$ length(concat(l_1, l_2)) = length(concat(nil, l_2)) =
  $\overset{ax1}{=}$ length(l_2) = length(l_2) + length(nil) = length(l_2) + length(l_1)
- ip. do ind length(concat(l, l_2)) = length(l) + length(l_2)
  $l_1 = cons(x, l)$
- dem. length(concat(cons(x, l), l_2)) = length(cons(x, l)) + length(l_2)

length(concat(cons(x, l), l_2)) $\overset{concat}{=}$

= length(cons(x, concat(l, l_2))) $\overset{length}{=}$

= 1 + length(concat(l, l_2)) $\overset{ip}{=}$

= 1 + length(l) + length(l_2) $\overset{length}{=}$

= length(cons(x, l)) + length(l_2)

factorial(n)

3

5 length(rev(l₁)) = length(l₁)

· caz de baza -> l₁ = nil => length(rev(l₁)) = length(rev(nil)) =
= length(nil) = length(l₁)

· ipoteza length(rev(l)) = length(l) ; l₁ = cons(x, l)

· dem length(rev(cons(x, l))) = length(cons(x, l))

length(rev(cons(x, l))) $\overset{rev}{=}$

= length(concat(rev(l), cons(x, nil))) $\overset{ex4}{=}$

= length(rev(l)) + length(cons(x, nil)) $\overset{ip}{=}$

= length(l) + 1 + length(nil) =

= length(l) + 1

= length(cons(x, l))

+ (n == 1)

6. $\forall l_1$, $rev(rev(l_1)) = l_1$

- caz de bază: $l_1 = nil \Rightarrow rev(rev(l_1)) = rev(rev(nil)) = rev(nil) = nil = l_1$
- ip de ind $rev(rev(l)) = l$: $l_1 = cons(x, l)$
- dem $rev(rev(cons(x, l))) = cons(x, l) = l_1$

$rev(rev(cons(x, l))) \overset{rev}{=}$

$= rev(concat(rev(l), cons(x, nil))) \overset{ex\ 5}{=}$

$= concat(rev(cons(x, nil)), rev(rev(l))) \overset{ip}{=}$

$= concat(rev(cons(x, nil)), l)) \overset{rev}{=}$

$= concat(concat(rev(nil), cons(x, nil)), l) =$

$= concat(concat(nil, cons(x, nil)), l) \overset{concat}{=}$

$= concat(cons(x, nil), l) \overset{concat}{=}$

$= cons(x, concat(nil, l))$

$= cons(x, l)$

$= l_1$

1.
```
factorial(n)
    if (n == 1)
        return 1;
    else
        return n * factorial(n-1);
```

$i = n \in \mathbb{N}$ $\quad$ $O = n \in \mathbb{N}^*$

$P_i(n): n \geq 1$ $\quad$ $P_o(n, r): r = n!$

Inductie:

- caz de bază
  $n = 1 \Rightarrow factorial(n) = factorial(1) = 1! = 1$
- ip de ind.
  $factorial(n-1) = (n-1)!$
- pas de inductie
  $factorial(n) = n * factorial(n-1) = n * (n-1)! = n!$

Se verifică inducția matematică $\Rightarrow$ partial corect.

2. plus(x, y)
if (x == 0)
return y;
else
return 1 + plus(x - 1, y)

$i : \mathbb{N} + \mathbb{N}$ $\quad 0 \cdot \mathbb{N}$

$P_i(x, y) = 1$ $\quad P_0((x, y), r)$ $\quad r = x + y$

Inductie:

- caz de baze:

$x = 0 \Rightarrow plus(0, y) = y = 0 + y$

- ipoteza de inductie

$plus(x - 1, y) = x - 1 + y$

- pas de inductie

$plus(x, y) = 1 + plus(x - 1, y) = 1 + x - 1 + y = x + y = plus(x, y)$

3. Fie G un graf neorientat cu n noduri ($n \geq 3$)
Urm. propr sunt echivalente
a) G conex, fără cicluri
b) G minimal fără cicluri
c) G minimal, conex

a → b
G conex ⇒ ∀ 2 noduri (u, v) sunt conectate printr-un drum
Dacă se adaugă o muchie între două noduri arbitrare (u, v) ⇒
⇒ apar 2 modalități de a ajunge de la u la v (u → v și u → ... → v) ⇒
⇒ se formează un ciclu ⇒ deci G nu are cicluri ⇒
⇒ G minimal, fără cicluri

b → c
G conex
Pp. că G nu este conex ⇒ ∃ 2 noduri (u, v) între care nu există drum ⇒
⇒ putem adăuga muchia (u, v), fără să se formeze un ciclu ⇒
⇒ G nu e minimal ⇒ contradicție ⇒ G - conex

G conex. minimal
Pp. că G nu e minimal, conex ⇒ se poate elimina muchia (u, v) și
graful rămâne conex ⇒ ∃ alt drum de la u la v ⇒ există cicluri
în graf ⇒ contradicție ⇒ G - minimal, conex

c → a
G minimal, conex ⇒ ∃ un drum unic (u, v) ⇒ dacă ar exista mai
mult de un drum între u și v ⇒ ar exista cicluri ⇒
⇒ G nu are cicluri

Struțu Ilinca - Joana 921 CC LAB 7

```
• find(A[][], x) {
    for(i=1; i<=m; i++)
        for(j=1; j<=m; j++)
            if(A[i][j] == x)
                return 1;
    return 0;
}
```

O(m*m)

```
• find(A[][], x) {
    j = choice({1, ..., m})
    for(i=1; i<=m; i++)
        if(A[i][j] == x)
            success;
    fail;
}
```

O(m)

```
• find(A[][], x)
    i = choice({1, ... m})
    j = choice({1, ... m})
    if(A[i][j] == x)
        success;
    fail;
}
```

P_1: $Q_2 \in P$
$Q_1 \leq_p Q_2$ $\not\Rightarrow Q_1 \in P$

P_2: $\leq_p$ este tranzitivă

(1) $Q_2 \in NP_\Delta \Leftrightarrow \exists\, Q_1 \in NP_C$ a.î. $Q_1 \leq_p Q_2$

"$\Rightarrow$" $Q_2 \in NP_\Delta \Rightarrow \forall Q' \in NP$ avem $Q' \leq_p Q_2$ $\not\Rightarrow Q_1 \leq_p Q_2$
$NP_C \subset NP$
$Q_1 \in NP_C$ $\not\Rightarrow Q_1 \in NP$

"$\Leftarrow$" $Q_1 \in NP_C \Rightarrow Q_1 \in NP_\Delta \Rightarrow \forall Q_3 \in NP,\ Q_3 \leq_p Q_1$
dar $Q_1 \leq_p Q_2 \Rightarrow Q_3 \leq_p Q_2$, $Q_2 \in P \Rightarrow Q_2 \in NP_\Delta$

(2) $\forall Q_1, Q_2 \in NP_C \Rightarrow Q_1 \leq_p Q_2$ și $Q_2 \leq_p Q_1$

$Q_1 \in NP_\Delta \Rightarrow Q_1 \in NP_\Delta \Rightarrow \forall Q_3 \in NP,\ Q_3 \leq_p Q_1$ $\not\Rightarrow$
$Q_2 \in NP_C \Rightarrow Q_2 \in NP$

$\Rightarrow Q_3 = Q_2 \leq_p Q_1$

Analog $Q_1 \leq_p Q_2$

(3) $NP_C \cap P \neq \emptyset \Rightarrow P = NP$

$Q \in NP_C \Rightarrow Q \in NP$ (1) și $Q \in NP_\Delta$ (1)

(1) $\Rightarrow \forall Q' \in NP \Rightarrow Q' \leq_p Q$, $Q \in P \Rightarrow Q' \in P$

$Q' \in P \Rightarrow \forall Q' \in NP \subset P$ deci $P \subseteq NP \Rightarrow P = NP$

(T4) $NP_Q \cap P \neq Q \Rightarrow P = NP$ $\quad Q' \leq_P Q \not\Rightarrow Q' \in P$

$Q \in NP_Q \Rightarrow \forall Q' \in NP \Rightarrow Q \in P$

$\Rightarrow \forall Q' \in NP \in P, P \subseteq NP = P = NP$

(C1) $NPC \cap P = \emptyset \Rightarrow NPC \subseteq P$

$\downarrow$

$P = NP$

$NPC \subseteq NP \not\Rightarrow NPC \subseteq P$

(C2) $NP_Q \cap P \neq \emptyset \Rightarrow NPC \subseteq P$

$\Rightarrow P = NP$

$NPC \subseteq NP \not\Rightarrow NPC \subseteq P$

2) A(G, k) → $O(m^3)$

- for $i = \overline{1,m}$ — $O(m^3)$
  - if B(G, i) → $O(m^2)$
    - break;
- if i = k
  - success
- fail

B(G, NC) → $O(m^4)$

- for $i = \overline{0, m-1}$
  - $c[i] = choice(\{1, \ldots, Nc\})$
- for_each $(x, y) \in E$
  - if $c[x] = c[y]$
    - fail
- success

```
TSP(V, L) {
    V' = ∅
    E = ∅
    for i = 1 ... n-1
        u = choice(V)
        if (u ∈ V')
            fail
        V' = V' ∪ {u}
    sum = 0
    V' = V' ∪ {1}
    for each i = 1 ... (n-1)
        sum = sum + V'[i] + V'[i+1].
    if sum < L
        success
    fail
}
```

Struțu Ilinca-Ioana

Laborator 9

① a)
```
fact1(n)
    r = 1
    for i=1:n
        r = r * i
    return r
```

Se face un singur apel și se alocă memorie pt. 2 variabile

$i \leq n \Rightarrow$ se folosește $O(\lg n)$

$r \leq n! \Rightarrow$ se folosește $O(\lg n!)$ $\Rightarrow$ Complexitatea spațială este $O(\lg n!)$

b)
```
fact2(n)
    if(n==0)
        return 1
    else
        return n * fact2(n-1)
```

Lungimea lanțului de recursivitate este n. O stivă va costa $O(\lg n)$ deoarece parametrul este $\leq n \Rightarrow$ Complexitatea spațială este $O(n \lg n)$.

② a)
```
suma1(v,n)   unde v[i]<n
    s = 0
    for i=1:n
        s = s + v[i]
```

Se face un singur apel și se folosește memorie pt. 2 variabile

$s \leq n \cdot n \Rightarrow O(\lg n \cdot n)$

$i < n \Rightarrow O(\lg n)$ $\Rightarrow$ Comp. sp. $O(\lg n \cdot n)$

b) suma2(v, i, j)

```
if i = j
    return v[i]
else
    return suma2(v, i, (i+j)/2) + suma2(v, (i+j)/2, j)
```

Lungimea lantului de recursivitate este $\lg n$

Pt. fiecare apel se foloseste memorie pt. 3 variabile

$i \leq n \Rightarrow O(\lg n)$

$j \leq n \Rightarrow O(\lg n)$

suma dim return $\leq n \cdot n \Rightarrow O(\lg(n \cdot n))$ $\Big/ \Rightarrow O(\lg n \cdot \lg n \cdot n)$

c) suma3(v, i, n)

```
if i = n
    return v[n]
else
    return v[i] + suma3(v, i+1, n)
```

Lungimea lantului de recursivitate este n

Pentru fiecare apel se foloseste memorie pt. 2 variabile:

$i \leq n \Rightarrow O(\lg n)$

suma dim return $\leq n \cdot n \Rightarrow O(\lg n \cdot n)$ $\Big/ \Rightarrow O(n \lg n \cdot n)$

③ N-TSP(G, l)

```
V' = ∅
for i = 1 : m
    v = choice(V)
    if v ∈ V'
        fail
    V' = V' ∪ {v}
s = 0
for i = 1 : (m-1)
    s = s + cost(v'[i], v'[i+1])
s = s + cost(v'[m], v'[1])
if s <= l
    success
fail
```

↳ variabile

V': m elem, $m \leq n \Rightarrow O(n \lg n)$

$i \leq m \Rightarrow O(\lg n)$

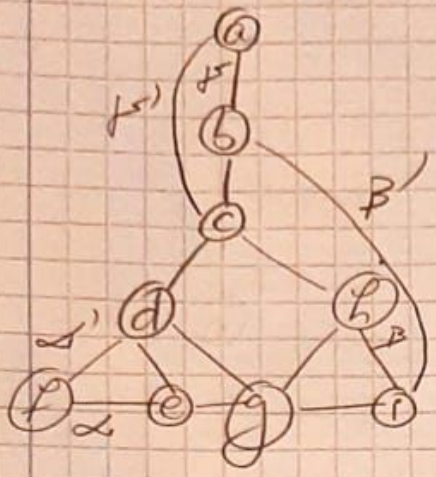
$s \leq n \cdot m \Rightarrow O(\lg nm)$

$v \leq m \Rightarrow O(\lg n)$

$\Rightarrow O(n \lg n + \lg nm)$

Stratu Bianca - Ioana

Laboratorul 10 - 13/01/2021 (curs 16)



Să se rulbreze pe foaie alg. asc_2_conectat(G) până la descoperirea punții ζ și imbricarea ei cu arcul invers ζ'

	c	π	b	b_2	d	e
a	∅ 1	∅	1	1	1	(a,a)
b	∅ 1	a	2	2	2	(b,b)
c	∅ 1	b	3	3	3	(c,c)(c,a)
d	∅ 1	c	4	4	4	(d,d)
e	∅ 1	d	5	5	5	(e,e)(f,d)
f	∅ 2	e	6 4	6	6	(f,f)(f,d)
g	∅					
h	∅					
i	∅					

asc_2_conectat(G)

c(a..i) = a|b

π(a..i) = null

timp = 0

E_2 = ∅

- explorare(a)

d(a) = b(a) = b_2(a) = 1

e(a) = (a,a); c(a) = conținu

succesori: b, c

- b

c(b) = a|b Ⓐ => π(b) = a; $E_2 = E_2 \cup \{(a,b)\}$

- explorare(b)

d(b) = b(b) = b_2(b) = 2

e(b) = (b,b); c(b) = conținu

succesori: c

- c

c(c) = alb (A) => π(c) = b; $E_2 = E_2 \cup \{(b,c)\}$

• explorare(c)

d(c) = b(c) = b_2(c) = 3

e(c) = (c,c)

succesori: a, d, b

- a

c(a) = alb (F)

π(a) ≠ c (A) => $d_A(a) < b_3(c)$ (A) =>

=> b(c) = d(x) = 1

e(c) = (c,a)

- d

c(d) = alb (A) => π(d) = c; $E_2 = E_2 \cup \{(c,d)\}$

• explorare(d)

d(d) = b(d) = b_2(d) = 4

c(d) = comusica

e(d) = (d,d)

succesori: e, f

- e

c(e) = alb (A) => π(e) = d; $E_2 \cup \{(d,e)\}$

• explorare(e)

d(e) = b(e) = b_2(e) = 5

c(e) = comusica

e(e) = (e,e)

succesori f, g

- f

$c(f) = alb$ (A) $\Rightarrow \pi(f) = e$, $E_1 = E_2 \cup \{(e, f)\}$

- explorare f

$d(f) = b(f) = b_2(f) = 6$

$e(f) = (f, f)$

$c(f) =$ conuzie

succesori: d

- d

$c(d) = alb$ (F) $\Rightarrow \pi(f) \neq d$ (A) $\Rightarrow d_4(d) < b_6(f)$ (A) $\Rightarrow$

$\Rightarrow b(f) = d(d) = 4$

$e(f) = (f, d)$

$c(f) =$ negru

$b_4(f) < b_5(e)$ (A) $\Rightarrow$

$\Rightarrow b(e) = b(f) = 4$

$e(e) = e(f) = (f, d)$

$d_5(e) < b_6(f)$ (A) $\Rightarrow$

$\Rightarrow E_2 = E_2 \cup \{e(f)\}$

Ne oprim pentru că muchia $a = (e, f)$ este acoperită de arcul invers $a' = (f, d)$